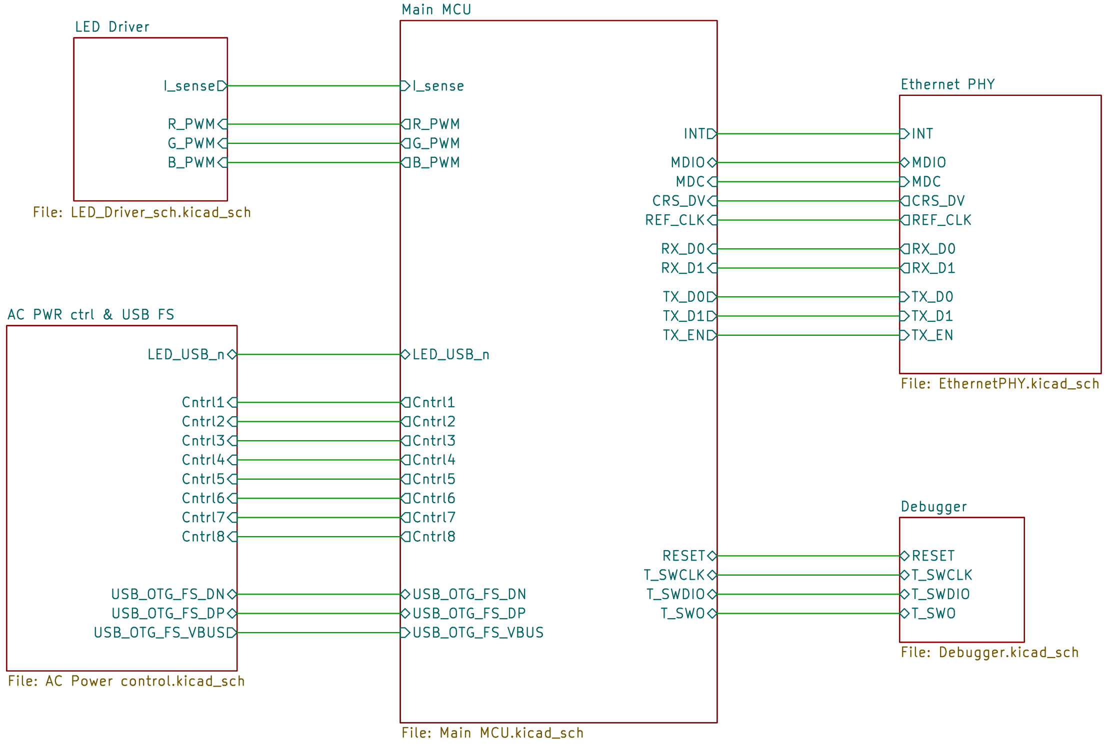

For reference only

Block Diagram

Power Budget

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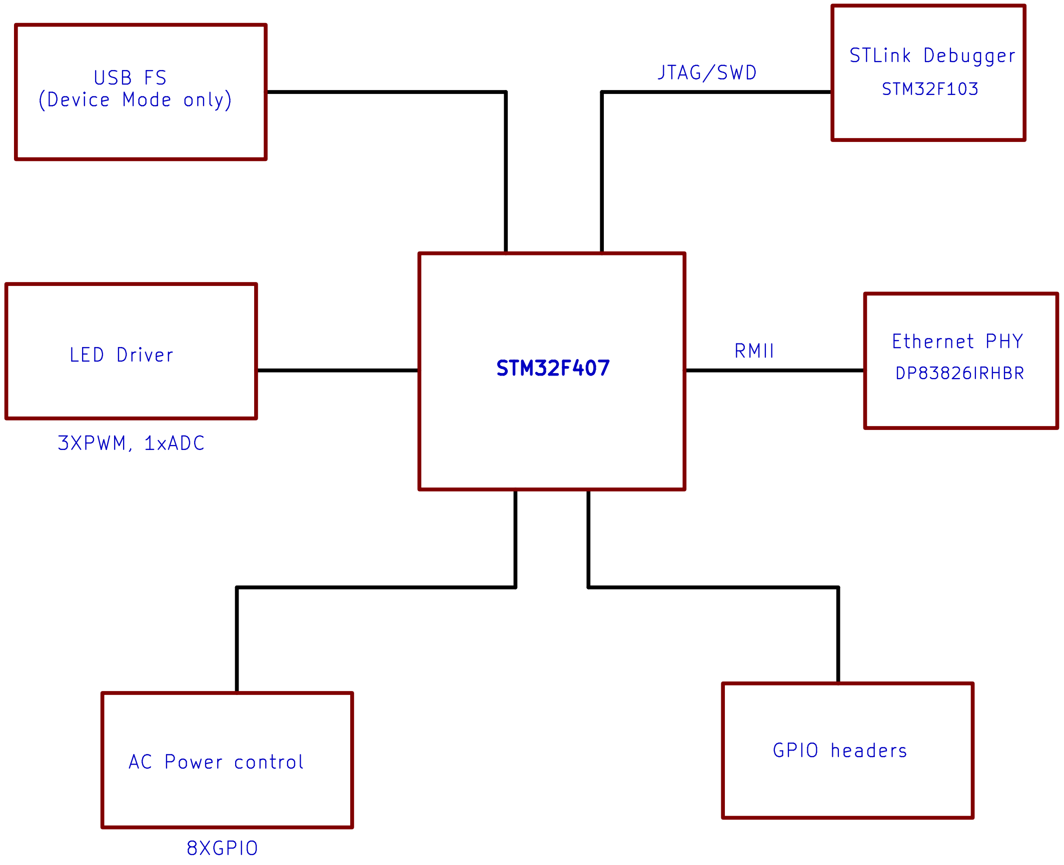
Sheet: /
File: STM32 Microcontroller Board.kicad_sch

Title: DeskNode-F4

Size: A4 | Date: | **Rev: A**

KiCad E.D.A. 9.0.6 | Id: 1/8



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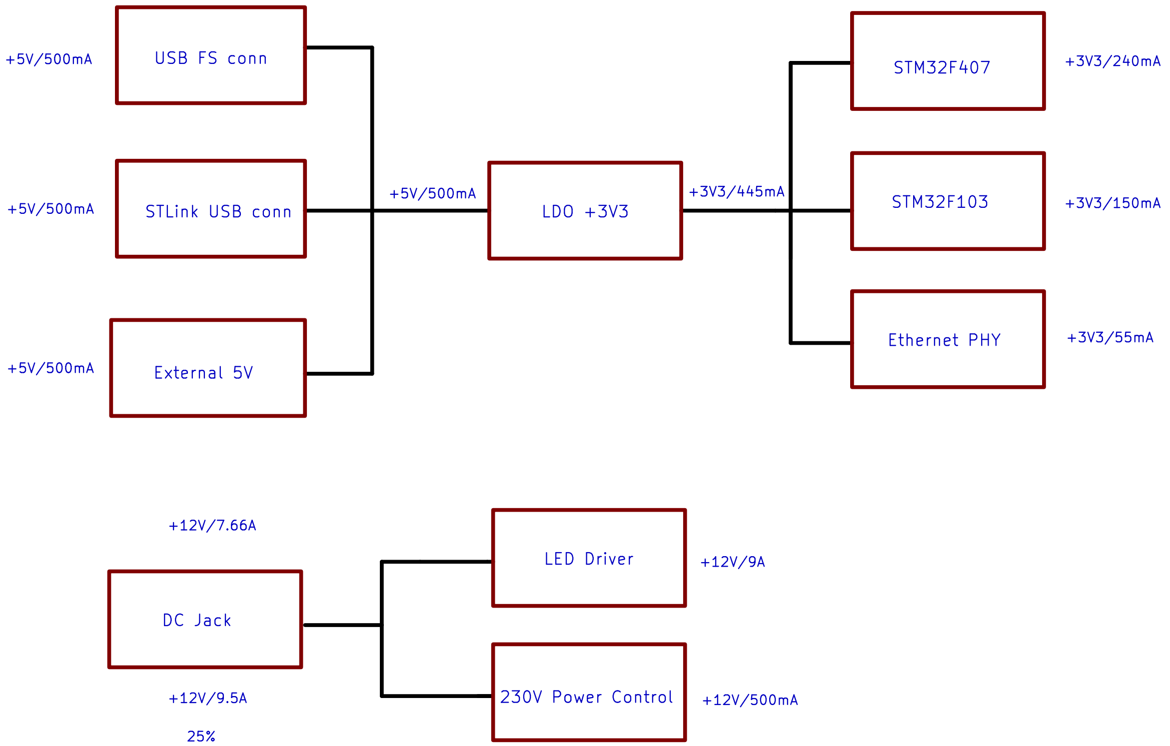
Sheet: /Block Diagram/
File: Block Diagram.kicad_sch

Title: DeskNode-F4

Size: A4	Date:	**Rev: A**
KiCad E.D.A. 9.0.6		Id: 2/8



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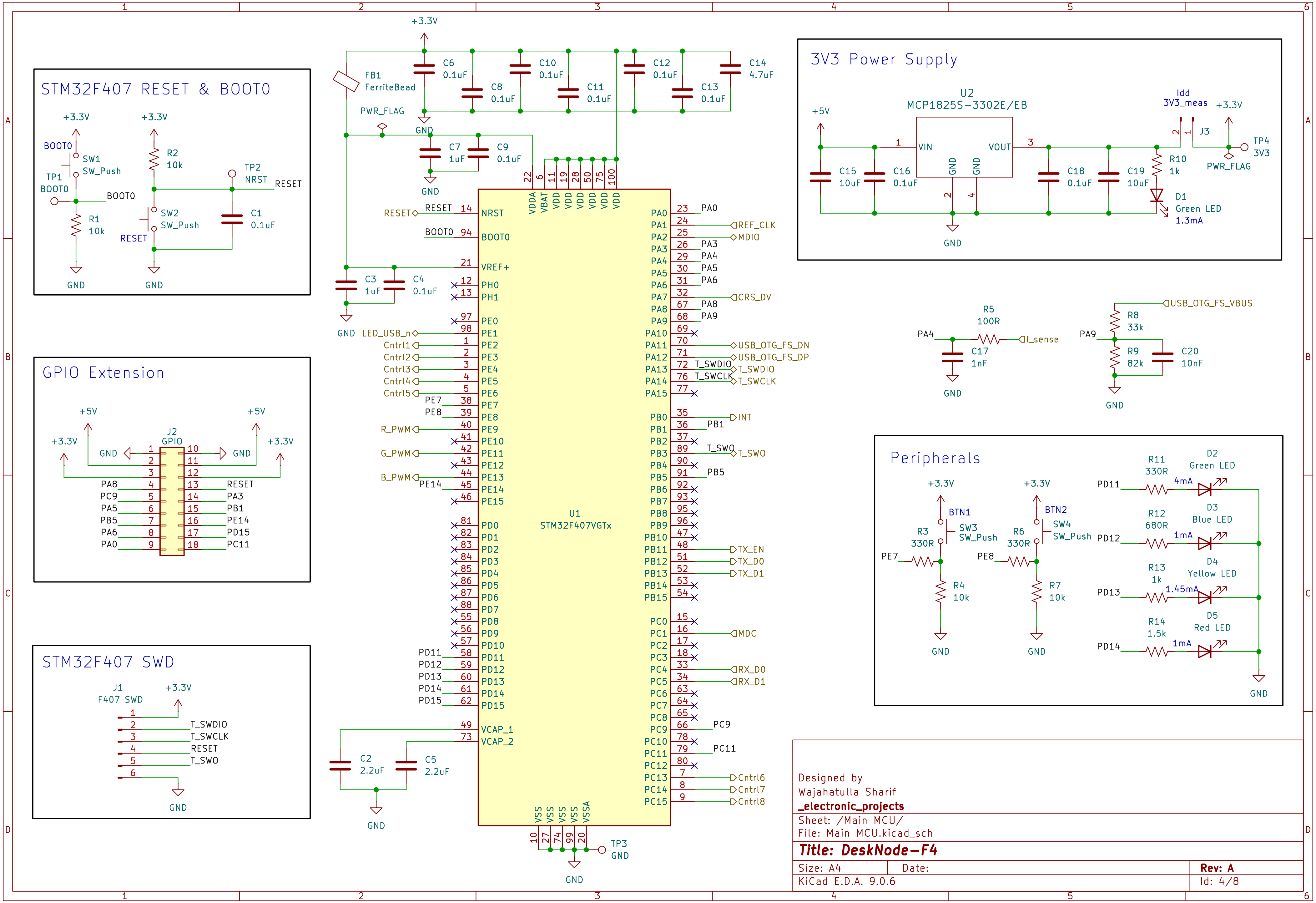
Sheet: /Power Budget/
File: Power Tree.kicad_sch

Title: DeskNode-F4

Size: A4	Date:	**Rev: A**
KiCad E.D.A. 9.0.6		Id: 3/8


STM32F407 RESET & BOOT0
+3.3V
BOOT0
SW1
SW_Push
TP1
BOOT0
R1
10k
GND
R2
10k
TP2
NRST
RESET
SW2
SW_Push
C1
0.1uF
GPIO Extension
+5V
+3.3V
J2
GPIO
GND
PA8
PC9
PA5
PB5
PA6
PA0
PA3
PB1
PE14
PD15
PC11
STM32F407 SWD
J1
F407 SWD
T_SWDIO
T_SWCLK
T_SWO
FB1
FerriteBead
PWR_FLAG
C6
0.1uF
C8
0.1uF
C10
0.1uF
C11
0.1uF
C12
0.1uF
C13
0.1uF
C14
4.7uF
C7
1uF
C9
0.1uF
C3
1uF
C4
0.1uF
C2
2.2uF
C5
2.2uF
U1
STM32F407VGTx
NRST
BOOT0
VREF+
VCAP_1
VCAP_2
VDDA
VBAT
VDD
VSS
VSSA
LED_USB_n
Cntrl1
Cntrl2
Cntrl3
Cntrl4
Cntrl5
Cntrl6
Cntrl7
Cntrl8
R_PWM
G_PWM
B_PWM
REF_CLK
MDIO
CRS_DV
USB_OTG_FS_DN
USB_OTG_FS_DP
INT
TX_EN
TX_D0
TX_D1
MDC
RX_D0
RX_D1
TP3
3V3 Power Supply
U2
MCP1825S-3302E/EB
VIN
VOUT
C15
10uF
C16
0.1uF
C18
0.1uF
C19
10uF
Idd
3V3_meas
J3
R10
1k
D1
Green LED
1.3mA
TP4
3V3
R5
100R
I_sense
C17
1nF
USB_OTG_FS_VBUS
R8
33k
R9
82k
C20
10nF
Peripherals
BTN1
SW3
SW_Push
R3
330R
R4
10k
BTN2
SW4
SW_Push
R6
330R
R7
10k
R11
330R
D2
Green LED
4mA
R12
680R
D3
Blue LED
1mA
R13
1k
D4
Yellow LED
1.45mA
R14
1.5k
D5
Red LED
1mA
Designed by
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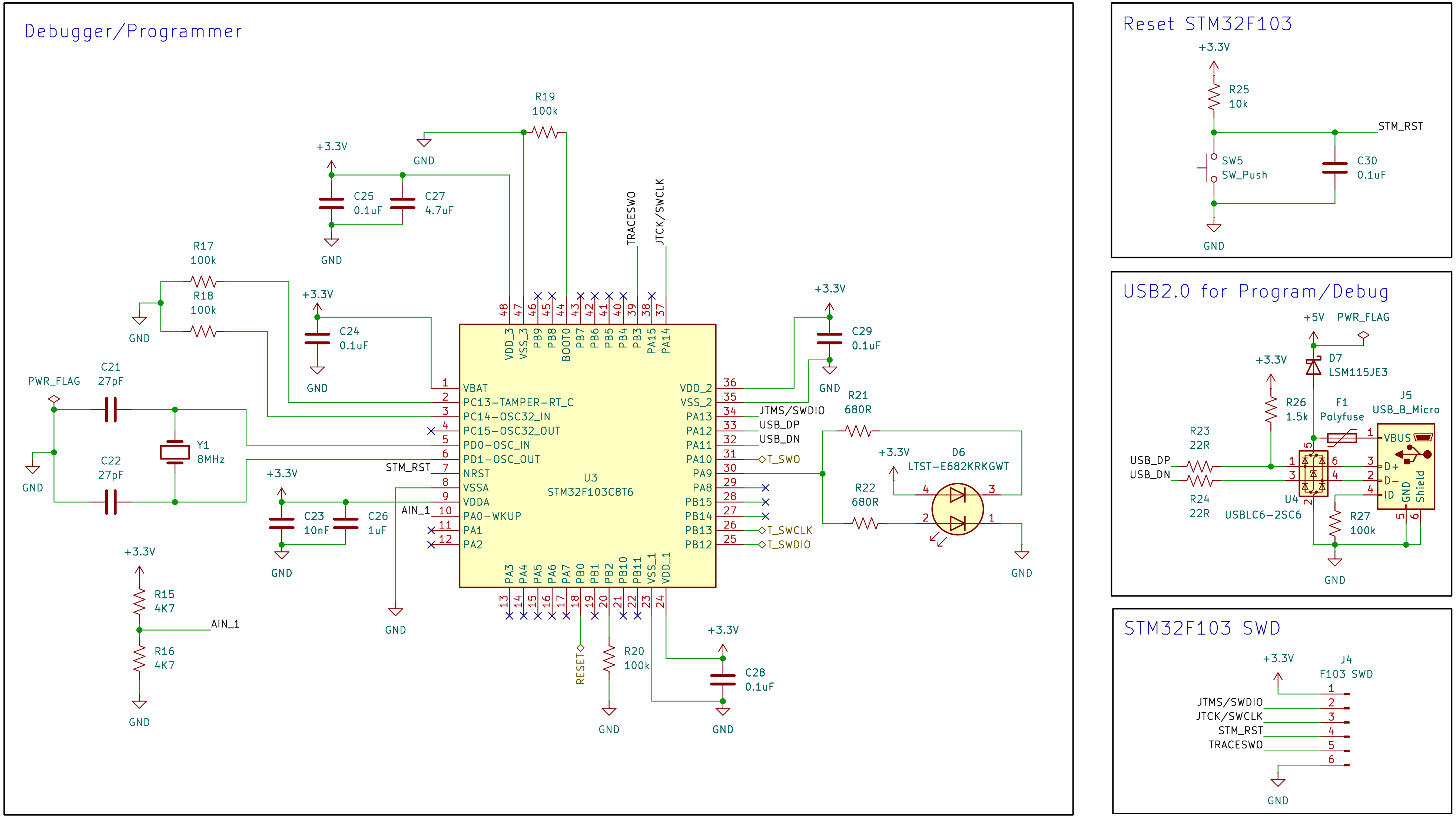
Sheet: /Main MCU/
File: Main MCU.kicad_sch
Title: DeskNode-F4
Size: A4
Date:
KiCad E.D.A. 9.0.6
Rev: A
Id: 4/8



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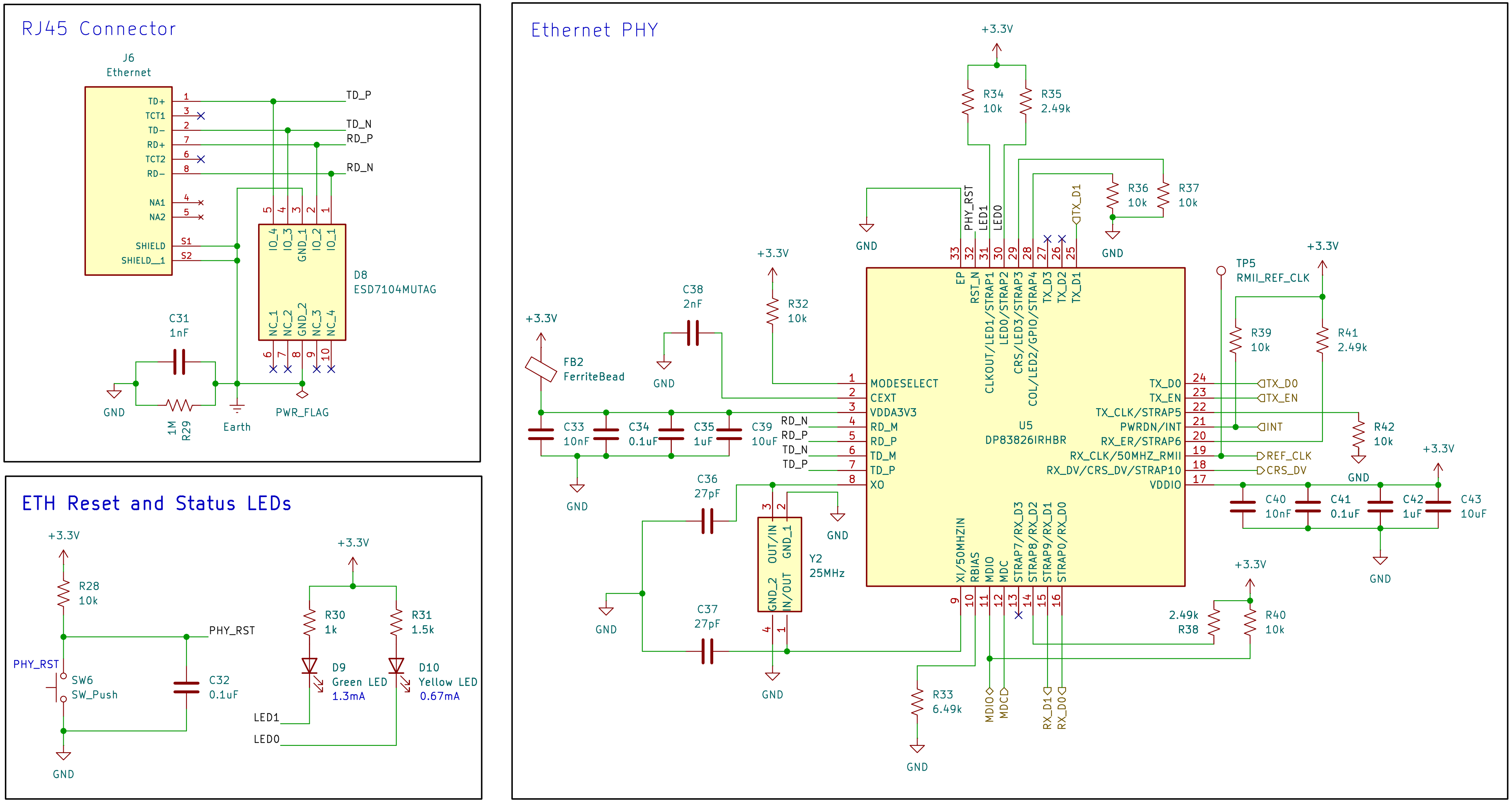
Sheet: /Debugger/
File: Debugger.kicad_sch

Title: DeskNode-F4

Size: A4	Date:	**Rev: A**
KiCad E.D.A. 9.0.6		Id: 5/8



PHY Bootstrap table

Pin Name	Strap	Pin No	Strap Fn	Default	Resistor	Final State
RX_D0	Strap0	16	Auto-negotiation en/dis	0	–	0 – enabled
LED1	Strap1	31	Old Nibble en/dis	1	10k	1 – enabled
LED0	Strap2	30	PHY_ADD0	0	2.49k	1
LED3	Strap3	29	PHY_ADD1	0	10k	0
LED2	Strap4	28	PHY_ADD2	0	10k	0
TX_CLK	Strap5	22	RMII Leader/Follower	0	10k	0 – Leader
RX_ER	Strap6	20	Pin 31 is CLKOUT 25MHz/LED1	0	2.49k	1 – LED1
RX_D3	Strap7	13	Pin 18 is CRS_DV/RX_DV	0	–	0 – CRS_DV
RX_D2	Strap8	14	MII/RMII	0	2.49k	1 – RMII
RX_D1	Strap9	15	auto MDIX en/dis	0	–	0 – enabled
RX_DV	Strap10	18	MDIX/MDI (only applicable if auto-MDIX is dis)	0	–	0 – NA

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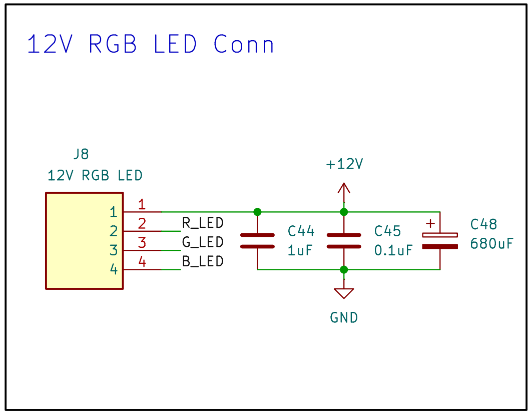
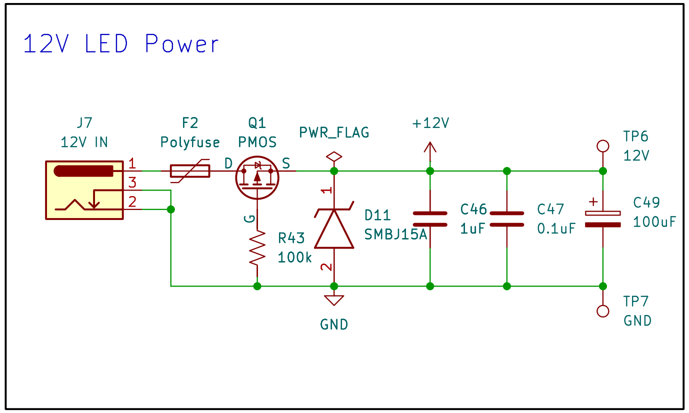
Sheet: /Ethernet PHY/
File: EthernetPHY.kicad_sch

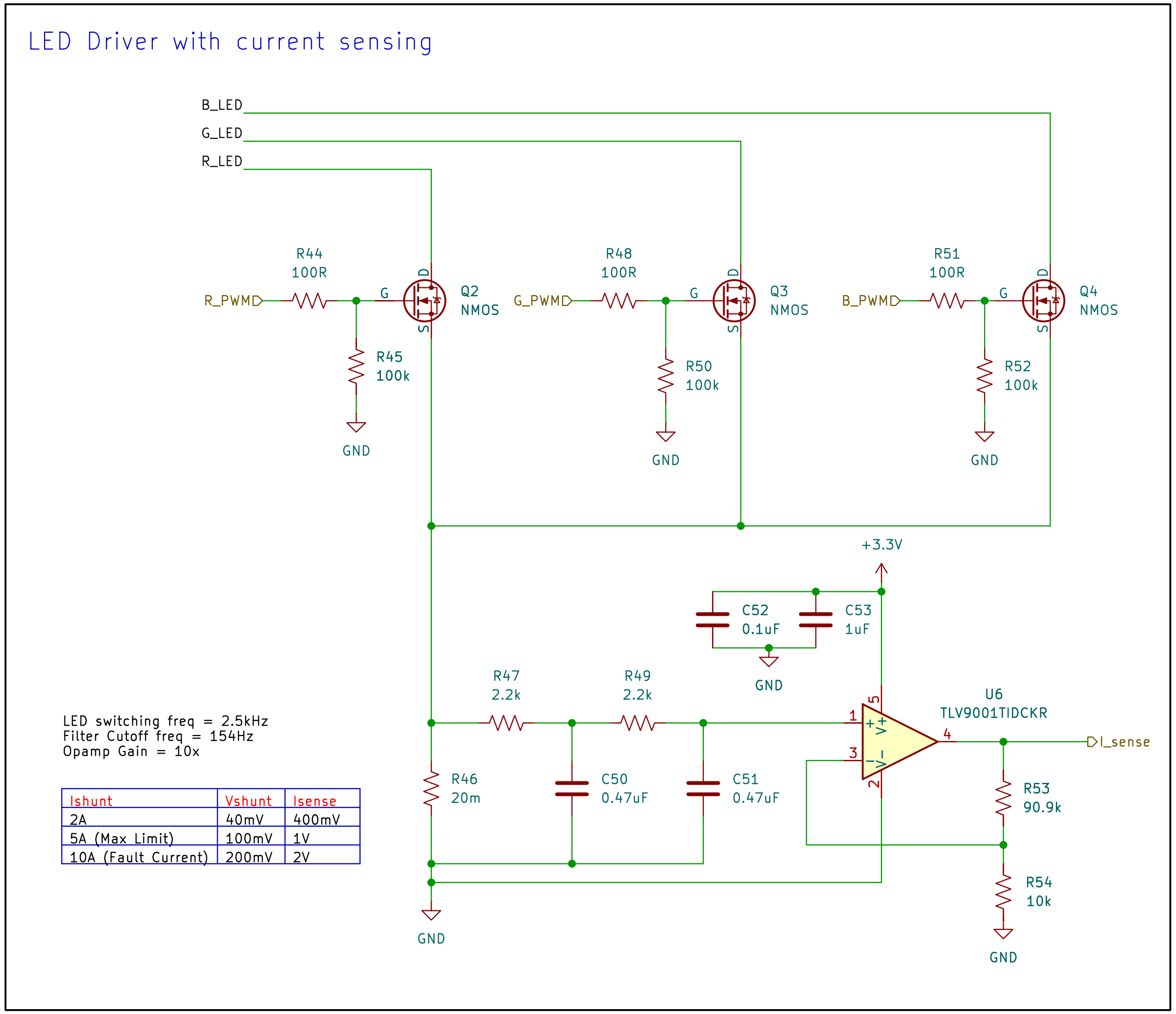
Title: DeskNode-F4

Size: A4 | Date: | **Rev: A**

KiCad E.D.A. 9.0.6 | Id: 6/8







LED switching freq = 2.5kHz
Filter Cutoff freq = 154Hz
Opamp Gain = 10x

Ishunt	Vshunt	Isense
2A	40mV	400mV
5A (Max Limit)	100mV	1V
10A (Fault Current)	200mV	2V

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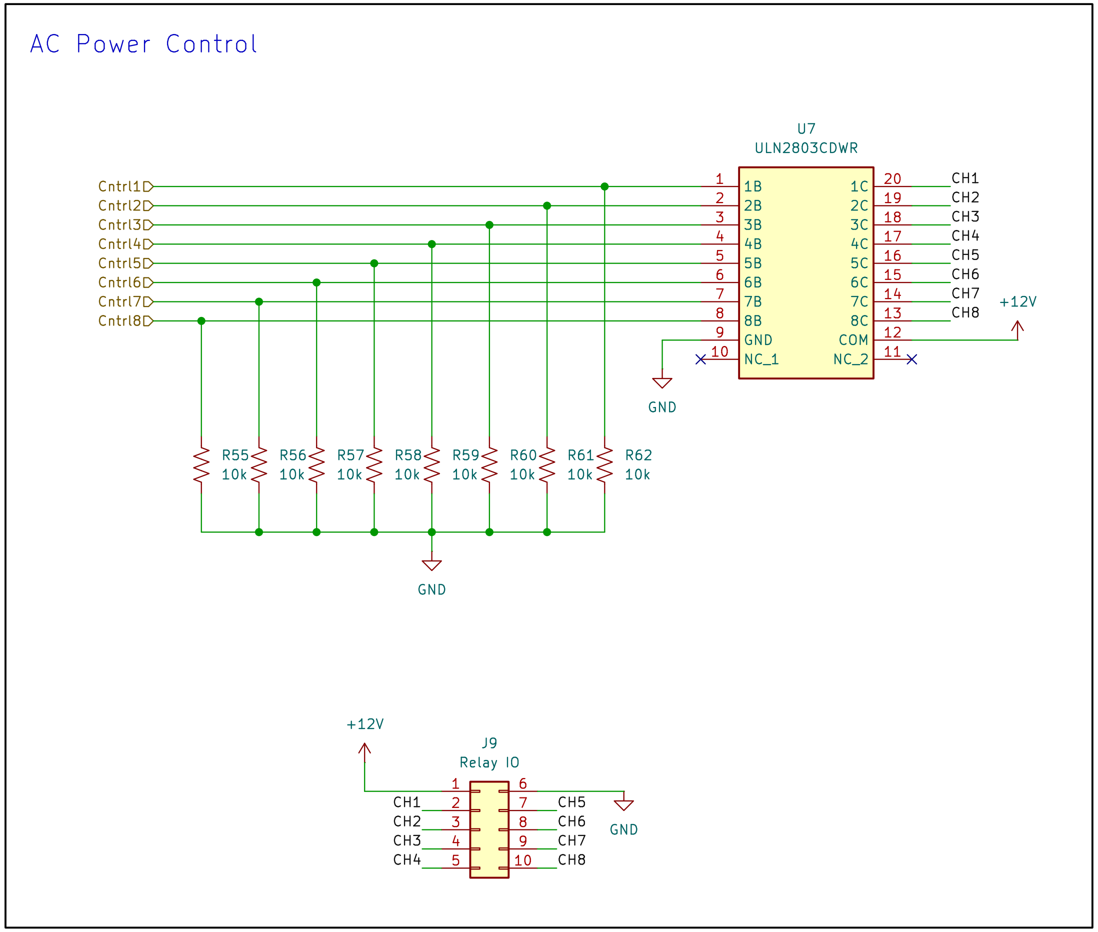
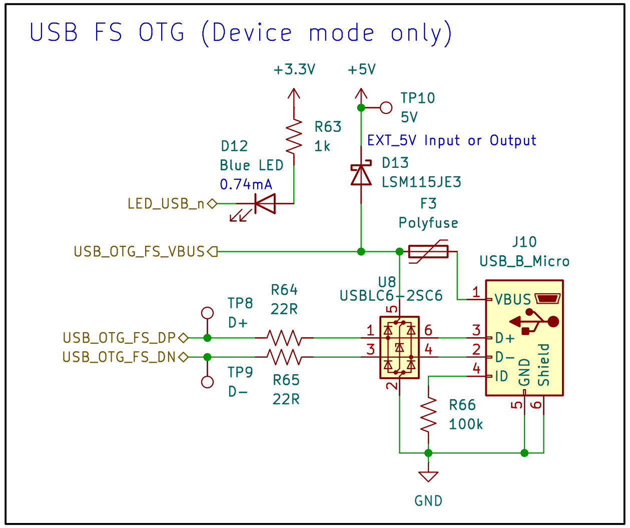
Sheet: /LED Driver/
File: LED_Driver_sch.kicad_sch

Title: DeskNode-F4

Size: A4 Date: **Rev: A**
KiCad E.D.A. 9.0.6 Id: 7/8





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Sheet: /AC PWR ctrl & USB FS/
File: AC Power control.kicad_sch

Title: DeskNode-F4

Size: A4 | Date: | **Rev: A**

KiCad E.D.A. 9.0.6 | Id: 8/8